WORKSHEET 9.2

A Phrase Bank for Justifications

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- **Extremum**: "f' changes from positive to negative at $x = c$, so f has a relative maximum there." Name the sign change.
- **Theorems**: state the condition, then the conclusion. IVT and EVT need continuity; MVT needs continuity on $[a, b]$ and differentiability on (a, b).
- **Over or under**: a tangent line lies below a graph that is concave up; a left sum underestimates for an increasing function.
- **Speed**: increasing when v and a have the same sign. Give both values.

Worked examples

Example 1 · A sign change

$f'(x) = (x - 2)(x + 1)^2$. Justify the location of every relative extremum of f.

Solution. f' changes from negative to positive at $x = 2$. At $x = -1$ it is negative on both sides.

A relative minimum at $x = 2$ only; no extremum at $x = -1$ because f' does not change sign.

Example 2 · A theorem

f is continuous on $[1, 5]$ with $f(1) = -3$ and $f(5) = 4$. Must $f(c) = 0$ for some c in $(1, 5)$?

Solution. 0 lies between $f(1)$ and $f(5)$.

Yes: f is continuous on $[1, 5]$ and $-3 < 0 < 4$, so by the Intermediate Value Theorem such a c exists.

Example 3 · Over or under

$f''(x) > 0$ for all x. Is a tangent-line approximation of f an overestimate or an underestimate?

Solution. The graph is concave up, so it lies above each tangent line.

An underestimate, because $f'' > 0$.

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Practice

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CORE

1. $f'(3) = 0$ and $f''(3) = -2$. What can you conclude, and why?

2. $g'(x) > 0$ for all x in $(0, 4)$. Write a one-sentence justification that g is increasing on $[0, 4]$.

3. h is differentiable on $[2, 6]$ with $h(2) = 1$ and $h(6) = 13$. What does the Mean Value Theorem guarantee?

4. Is a left Riemann sum for a decreasing function an overestimate or an underestimate of the integral?

EXAM LEVEL

5. f is differentiable for all x, with $f(0) = 5$, $f(2) = 1$ and $f(5) = 7$. Explain why $f'(c) = 0$ for some c in $(0, 5)$.

6. $f''(x) = (x-1)^2(x-4)$. Find the x-coordinate of every point of inflection of f, with justification.

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7. $f''(x) > 0$ on $[a, b]$. Is a trapezoidal sum for $\int_a^b f(x)\,dx$ an overestimate or an underestimate? Give a reason.

8. A particle has $v(3) = -4$ and $a(3) = 2$. Is its speed increasing or decreasing at $t = 3$? Give a reason.

Question 9 · Free-response practice

The function f is twice differentiable. Selected values of f and f' are given in the table.

x	1	3	5	8
$f(x)$	2	6	5	11
$f'(x)$	3	1	-2	4

(a) Must there be a value c, $3 < c < 5$, with $f'(c) = -\frac{1}{2}$? Justify your answer.

(b) Must there be a value c, $5 < c < 8$, with $f'(c) = 0$? Justify your answer.

(c) Use the tangent line at $x = 3$ to approximate $f(3.5)$. Given that $f''(x) < 0$ for $3 < x < 5$, is the approximation an overestimate or an underestimate? Give a reason.

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Worked solutions

1. f has a relative maximum at $x = 3$, by the Second Derivative Test: $f'(3) = 0$ and $f''(3) < 0$.
2. g is increasing on $[0, 4]$ because $g'(x) > 0$ on $(0, 4)$.
3. h is differentiable, hence continuous, so there is a c in $(2, 6)$ with $h'(c) = \dfrac{13 - 1}{6 - 2} = 3$.
4. An overestimate: each rectangle uses the left, largest, value.
5. f is continuous, and 5 lies between $f(2) = 1$ and $f(5) = 7$, so by the IVT $f(b) = 5$ for some b in $(2, 5)$. Then $f(0) = f(b)$, and the MVT on $[0, b]$ gives $f'(c) = 0$.
6. $x = 4$ only: f'' changes from negative to positive there. At $x = 1$, f'' is negative on both sides, so there is no change of concavity.
7. An overestimate: the graph is concave up, so each chord lies above the graph.
8. Decreasing, because $v(3) < 0$ and $a(3) > 0$ have opposite signs.
9. (a) Yes. f is differentiable, hence continuous, on $[3, 5]$, so by the MVT there is a c with $f'(c) = \dfrac{5 - 6}{5 - 3} = -\dfrac{1}{2}$.
   (b) Yes. f is twice differentiable, so f' is continuous on $[5, 8]$, and $f'(5) = -2 < 0 < 4 = f'(8)$. By the IVT, $f'(c) = 0$ for some c.
   (c) $f(3.5) \approx 6 + 1(0.5) = 6.5$. An overestimate: $f'' < 0$, so the graph is concave down and lies below its tangent line.

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